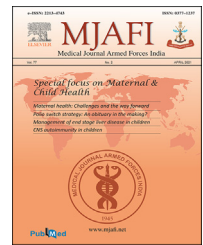


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Editorial

Maternal health: Challenges and the way forward

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Maternal health encompasses the healthcare dimensions of preconceptional care, antenatal care and obstetric care during childbirth as well as family planning and contraception to ensure safe motherhood. It also aims at making the entire experience of motherhood a fulfilling and positive experience. In other words, the healthcare is promotive, preventive, curative and rehabilitative for mothers. It has been a long-held measure of the health status of a country.

Sadly, despite such lofty ideals and objectives, maternal health is often ignored and not given its due importance at the national or even international level. The Millennium Development Goals were a step in the right direction and have greatly influenced the maternal health outcomes and mortality and morbidity from 2000 to 2015. The laid-down targets are yet to be met, especially in low-resource countries; however, there has been a perceptible decline in mortality and morbidity as far as maternal health is concerned.¹

New goals for global health have been set for 2030 by the UN General Assembly wherein Reproductive Maternal Newborn and Child health and Nutrition accounts for 13 of the Sustainable Development Goals for health.² Hence, the importance of maternal health has been recognised, but a comprehensive and multisectorial approach is needed to attain these goals at multiple community levels.

One would think these are problems of the third world or developing countries. However, the trend even in the most technologically advanced countries in the world raises concerns. The drop in maternal health mortality/morbidity has not been commensurate with the efforts in this direction. Even in the United States, the alarm bells rang at the highest echelons and resulted in the Department of Health and Human Services resolving to improve maternal health in an 'Action Plan' proposed in December 2020.³

The Action Plan noted that in 2018, the US maternal mortality rate was 17.4 deaths per 100,000 live births, with data

showing that as much as two-thirds of maternal deaths were preventable. Moreover, it is now acknowledged that for every chronicled maternal death, there are many more women who undergo serious complications at around the delivery time, which results in severe maternal morbidity.⁴

They have identified that the excessive caesarean rates especially in low-risk groups can result in complications, disability or even death. Another fact was that 33% of all pregnancy-related deaths were associated with, if not, attributed to, cardiovascular conditions.⁵ The Action Plan proposed to reduce the maternal mortality rate by 50% in 5 years, to reduce the low-risk caesarean rate by 25% and to achieve blood pressure control in 80% of women with hypertension.^{1,6}

India has come a long way over the last few decades as far as maternal health initiatives and health statistics go. The main aim of increasing institutional deliveries or at least having every birth attended by a skilled health assistant has been promoted by various government initiatives. The percentage of institutional births has doubled from 38.7% in 2005–2006 to 78.9% in 2015–2016 as per the National Family Health Survey. Unfortunately, this increase in coverage has not been reflected in a proportionate decrease in maternal or neonatal mortality and morbidity rates.

This most certainly points towards a few gaps in the quality of care provided by health facilities. The current national figures are 130 maternal deaths per 100,000 live births and 24 neonatal deaths per 1000 live births. Very few states such as Kerala and Goa have maternal mortality figures in double digits, 66 and 76 per 100,000 live births, respectively.

As mentioned earlier, approximately 46% maternal deaths and 40% stillbirths are preventable with higher quality healthcare. It is in this context that the government has launched the ambitious LaQshya programme in 2017, which aims to improve quality care in the labour rooms and

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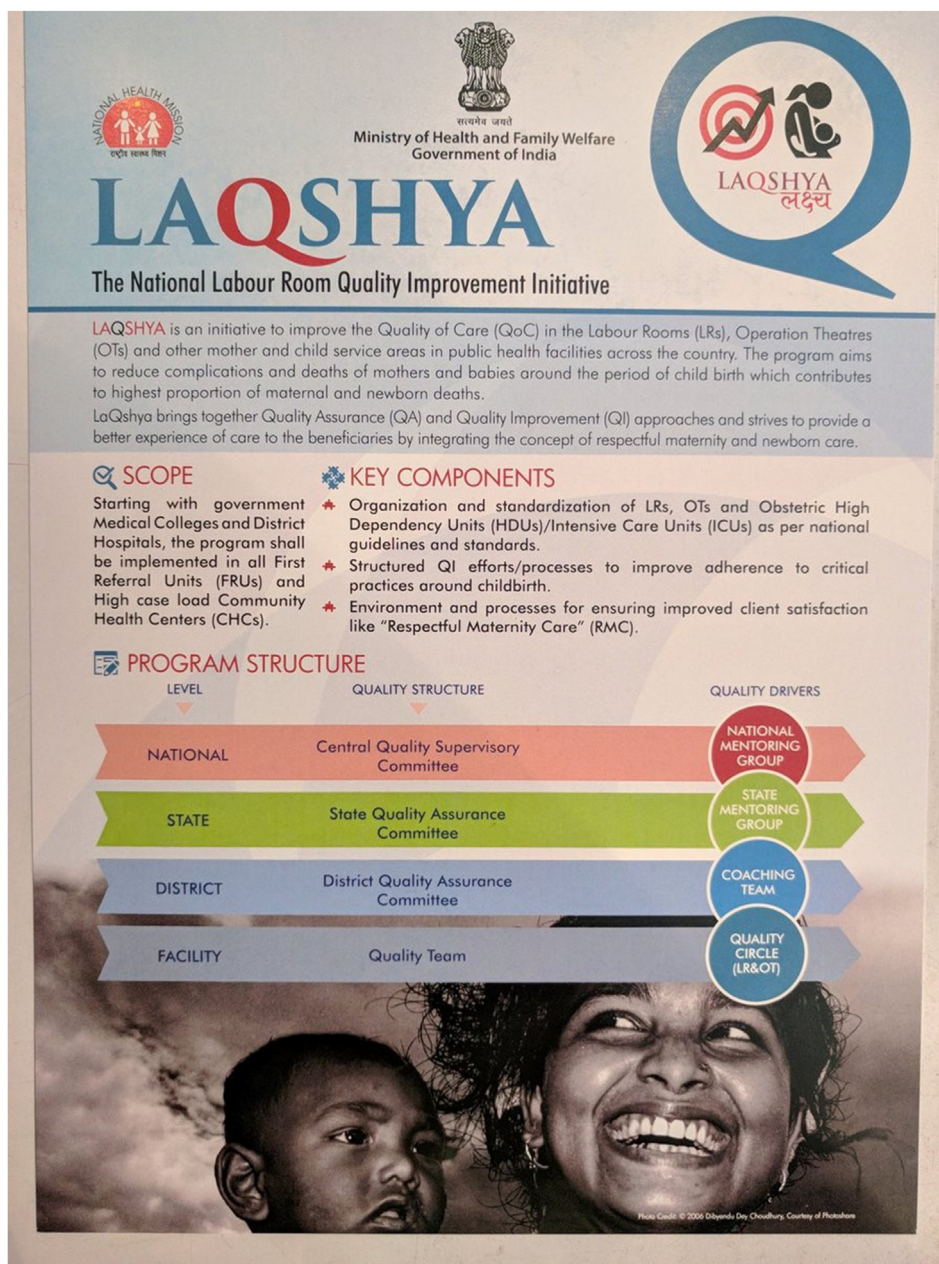


Fig. 1 – The LaQshya initiative.

maternity OT besides establishing and equipping obstetrics intensive care units and HDUs. A multipronged strategy aimed at upgradation of infrastructure, availability of essential equipment and capacity building of health resource persons is thought to improve the quality processes in labour rooms and OTs considerably. State-, district- and facility-level checks and balances have been put in place for mentoring and monitoring of these facilities and personnel (Fig. 1).

The LaQshya programme also has a web portal wherein all related data would be uploaded for prompt report generation and monitoring of progress. The facilities will also be assessed periodically, and any facility scoring higher than 70% on

quality assurance standards would be certified as a LaQshya-certified facility. They would be given platinum gold and silver badges as well as cash incentives to encourage best practices.

Maharashtra is a state that has endeavoured to go one step further and has launched the LaQshya-Manyata certification initiative for private facilities providing maternal and neonatal care.⁷ The staff and facility would be trained on 16 clinical and 10 facility standards espoused by the Health Ministry, and the private facility would be LaQshya-Manyata certified. This is worth replicating in other states as well because a sizeable chunk of antenatal and postnatal care is provided by the private sector.

Modalities to improve maternal health

Improve labour rooms

The Ministry of Health and Family Welfare, Govt of India (MoHFW) LaQshya document mentioned previously gives excellent methods of planning the new labour-delivery-recovery concept of labour beds, and its calculation depends on the number of deliveries per year. It also recommends new Labour delivery (LD) complexes to be built wherever required to meet these requirements. There are many hospitals that would benefit from these guidelines in establishing labour rooms that are client centric and assuring privacy, care and comfort during delivery. All healthcare institutes should aim to meet the LaQshya standards in every Labour Room (LR), make labour rooms LaQshya certified and endeavour to earn the badges as well.

The bedrock of ensuring safety and respectful maternal care would be the provision of well-equipped (both human resources and infrastructure and equipment) labour rooms, eminently outlined in the LaQshya guidelines (<http://www.nhm.gov.in/images/pdf/programmes/maternal-health/guidelines/Labor-Room%-20 Guideline.pdf>).

Training

The second most critical area as brought out by the global and national experience is the training of personnel, and this includes the entire team including gynaecologists, nursing officers and medical officers to become skilled healthcare givers, ensuring quality services to patients.

It is essential that our postgraduate residents be equipped with the knowledge of setting up and running or maintaining a well-equipped labour room unit as laid down in various quality assurance documents. This training should be part of the curriculum in Post Graduate (PG) training as they are most likely going to function as independent specialists immediately after postgraduation. Periodic evaluation and audits need to be carried out diligently to assess the efficacy of these training modules.

Ancillary and supportive measures

Maternal health is inextricably linked to good family planning or contraceptive practices and strategies. It is imperative that the latest long-acting reversible methods of contraception be included as a part of the armamentarium to give the clientele the choice of limiting the families to the desired size. Newer, safer and effective methods such as injectables and implants should be made widely available to the clientele at the earliest.

In providing optimum maternal health, supporting services such as radiology, blood bank availability, good neonatal support and critical care facilities all contribute substantially. In many of the smaller hospitals and clinics, one or many of these facilities are unavailable although they function as a centre of obstetric care. This needs rectification.

Audits and checks

Audits and checks are already in place in our system for maternal mortality cases; however, we need to make a full-fledged effort to make sure that self-auditing as well as auditing by competent authorities are in place whenever there is a 'near miss' or severe morbidity as well.⁸ Audits of the Cesarean Section (C-Sec) rates in various hospitals should also be carried out, taking into consideration the level of care they are involved in and the type of clientele they are dealing with.

Vaccination

This is another cornerstone of maternal health.⁹ This would include vaccinating adolescent or unmarried girls against rubella alongside guarding them against carcinoma of the cervix by using appropriate vaccination at appropriate times.

Adolescent health

This should be given utmost importance, and patient education should be a constant process in the community including schools and other academic and welfare institutions. Screening for adolescent Poly Cystic Ovarian Syndrome (PCOS), screening for obesity and spreading awareness among them regarding safe practices, menstrual hygiene and rights to maternal health are imperative.

The way forward

The coverage levels of life-saving interventions including modern methods of family planning, all births to be attended by a skilled attendant and vaccination should be universal, bridging the inequities between rich and poor and between ethnic groups and social status.

A major area of concern is the gaps in the quality of care delivered at health facilities. The interventions put in place may not result in the expected health results. Hence, studies are required to be conducted on the impact of essential inputs (diagnostics, medicines and equipment) on the contents of the care provided (data of care provided by healthcare facilities) and on the outcomes and impact (effective coverage, reduction in mortality and morbidity rates).¹⁰

It is essential to understand that several interventions starting from awareness of health rights served among the population to the availability of family planning programmes even to adolescents and to all women are irretrievably inter-linked. Basic guidelines needed for safe and effective delivery care across all healthcare facilities (all government/trusts/private/corporate sectors) should be laid down, and adherence to these minimum quality standards must be ensured, on fear of penalty, if need be. Availability of emergency obstetric care expressed as a percentage (5 facilities per 500,000 population) is essential for the health of every pregnant woman.¹¹

There is also a great need to conduct comprehensive analyses and surveys to measure the impact of interventions

used. A good way to identify and analyse is the three delays method wherein delays in deciding to seek care, reaching the healthcare facility and receiving care are analysed.¹²

The way forward is a multisectorial approach including governance, public awareness, policymaking and implementation of and strengthening the health sector at every level of public and private health delivery systems.

REFERENCES

1. World Health Organization. *Trends in Maternal Mortality 2000 to 2017: Estimated by WHO, UNICEF, UNFPA, World Bank Group and the United Nations Population Division*; 2019.
2. United Nations. *Transforming Our World: The 2030 Agenda for Sustainable Development*. Resolution A/RES/70/1 Adopted September 2015.
3. The Surgeon General's Call to Action to Improve Maternal Health Dec 2020. <https://www.hhs.gov/sites/default/files/callto-action-maternal-health.pdf>.
4. Office of Disease Prevention and Health Promotion, U.S. Department of Health and Human Services. Maternal, Infant, and Child Health. <https://www.healthypeople.gov/2020/topics-objectives/topic/maternalinfant-and-child-health>.
5. Centers for Disease Control and Prevention. Pregnancy Mortality Surveillance System. <https://www.cdc.gov/reproductivehealth/maternal-mortality/pregnancymortality-surveillance-system.html>.
6. Zaharatos J, St Pierre A, Cornell A, Pasalic E, Goodman D. Building U.S. capacity to review and prevent maternal deaths. *J Wom Health*. 2018;27(1):1–5.
7. What Is LaQshya- Manyata Program. http://www.maharashtralaqshyamanayata.org/wp-content/uploads/2019/07/What-is-LaQshya_Manyata.pdf.
8. England N, Madill J, Metcalfe A, et al. Monitoring maternal near miss/severe maternal morbidity: a systematic review of global practices. *PLoS One*. 2020;15(5), e0233697. <https://doi.org/10.1371/journal.pone.0233697>. PMID: 32470099; PMCID: PMC7259583.
9. Fortner KB, Nieuwoudt C, Reeder CF, Swamy GK. Infections in pregnancy and the role of vaccines. *Obstet Gynecol Clin N Am*. 2018;45(2):369–388. <https://doi.org/10.1016/j.ogc.2018.01.006>. PMID: 29747736.
10. Countdown to 2030 Collaboration. Countdown to 2030: tracking progress towards universal coverage for reproductive, maternal, newborn, and child health. *Lancet*. 2018;391(10129):1538–1548. [https://doi.org/10.1016/S0140-6736\(18\)30104-1](https://doi.org/10.1016/S0140-6736(18)30104-1). Epub 2018 Jan 31. PMID: 29395268.
11. Pattinson RC, Bergh AM, Ameh C, et al. Reducing maternal deaths by skills-and-drills training in managing obstetric emergencies: a before-and-after observational study. *S Afr Med J*. 2019;109(4):241–245. <https://doi.org/10.7196/SAMJ.2019.v109i4.13578>. PMID: 31084689.
12. Mohammed MM, El Gelany S, Eladwy AR, et al. A ten year analysis of maternal deaths in a tertiary hospital using the three delays model. *BMC Pregnancy Childbirth*. 2020;20(1):585. <https://doi.org/10.1186/s12884-020-03262-7>. PMID: 33023523; PMCID: PMC7541230.